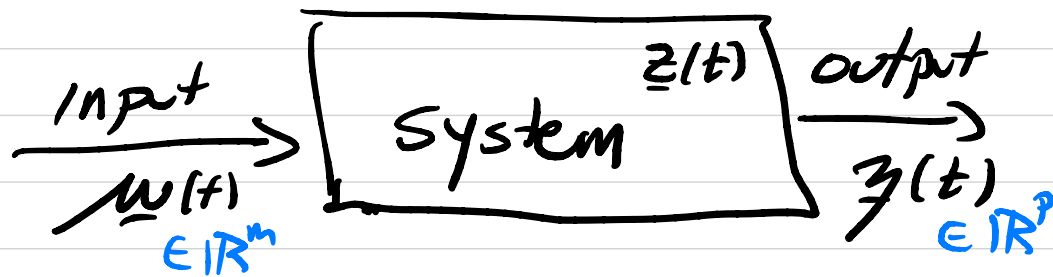


The state of a system is a collection of quantities $z_1, z_2, \dots, z_n$ such that knowledge of these quantities at a single time, t_0 , together with knowledge of the inputs $\underline{u}(t)$, $t \geq t_0$, is sufficient to predict the outputs $\underline{y}(t)$ $\forall t \geq t_0$



$$\{ \underline{z}(t_0), \underline{u}(t) \, t \geq t_0 \} \longrightarrow \underline{y}(t), t \geq t_0$$

feasible mapping
must make constructive

$$\underline{z}(t) = \begin{bmatrix} z_1(t) \\ z_2(t) \\ \vdots \\ z_n(t) \end{bmatrix} \in \mathbb{R}^n$$

state vector of system

Assume physical modeling specifies rate of change of
Chosen states $z_i(t)$, via implicit DEs of form

$$\begin{cases} 0 = f_k(\dot{z}_1(t), \dot{z}_2(t), \dots, \dot{z}_n(t), z_1(t), \dots, z_n(t), u_1(t), \dots, u_m(t), t) \\ k=1, \dots, n \quad (n \text{ eq'ns}) \end{cases}$$

i.e. rate of change of each $z_k(t)$ potentially depends
on the instantaneous values of the n states, the n
state derivatives, the m inputs, and time t itself
($2n+m+1$ variables).

More compactly:

$$\begin{cases} 0 = f_k(\dot{\underline{z}}(t), \underline{z}(t), \underline{u}(t), t) \\ k=1, \dots, n \end{cases}$$

Even more compactly, let

$$\underline{f}(\dot{\underline{z}}, \underline{z}, \underline{u}, t) = \begin{bmatrix} f_1(\dot{\underline{z}}, \underline{z}, \underline{u}, t) \\ f_2(\dot{\underline{z}}, \underline{z}, \underline{u}, t) \\ \vdots \\ f_n(\dot{\underline{z}}, \underline{z}, \underline{u}, t) \end{bmatrix} \quad (\text{vector of functions})$$

Then

$$\underline{\dot{Q}} = \underline{f}(\dot{\underline{z}}(t), \underline{z}(t), \underline{u}(t), t)$$

is the state equation for the system

Note: common alternate form is the explicit form

$$\dot{\underline{z}}(t) = \underline{f}(\underline{z}(t), \underline{u}(t), t)$$

But this can be an unnecessarily restrictive assumption
in some cases.

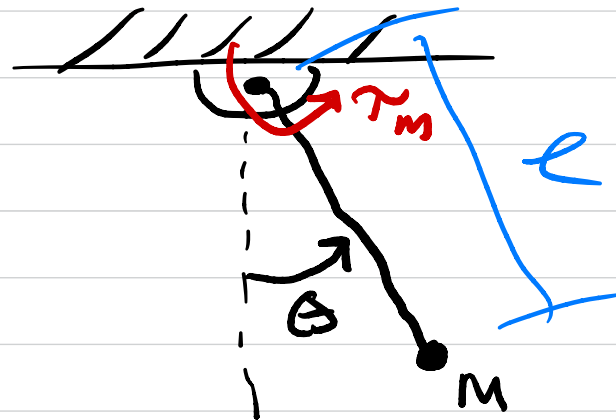
The state equation is linear, if:

$$\underline{f}(\dot{\underline{z}}, \underline{z}, \underline{u}, t) = \underbrace{E(t)}_{n \times n} \dot{\underline{z}} + \underbrace{F(t)}_{n \times n} \underline{z} + \underbrace{G(t)}_{n \times m} \underline{u}$$

Example:

Simple pendulum

Take $z_1 = \Theta$
 $z_2 = \dot{\Theta} = \omega$



Physics:

$$m l^2 \dot{\omega} = -m g l \sin \Theta - \underbrace{b(\omega)}_{\text{friction, drag}} + \underbrace{\tau_m}_{\text{Applied torque (servo motor)}} \sim \text{input, } u$$

$$\dot{\Theta} = \omega$$

Eqn:

$$\begin{aligned} \dot{z}_1 &= z_2 \\ m l^2 \dot{z}_2 &= -m g l \sin z_1 - b(z_2) + u \end{aligned}$$

Rewrite:

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{z}_1 - z_2 \\ m l^2 \ddot{z}_2 + m g l \sin z_1 + b(z_2) - \omega \end{bmatrix} \sim \begin{bmatrix} f_1(\dot{z}, z, \omega, t) \\ f_2(\dot{z}, z, \omega, t) \end{bmatrix}$$

$$\underline{0} = \underline{f}(\underline{\dot{z}}, \underline{z}, \omega, t)$$

Note that this is not linear, b/c of $\sin z_1$ term, and b/c $b(z_2)$ is not explicitly specified

However, if we assume $\sin z_1 \approx z_1$, $b(\omega) \approx \beta \omega$ (linear frict)

Then state eq'n is linear w/tn

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & m l^2 \end{bmatrix}}_E \begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & -1 \\ m g l & \beta \end{bmatrix}}_F \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ -1 \end{bmatrix}}_G \omega$$

(Note: E, F, G constant $\Rightarrow$ eq'n is also time invariant)

"Standard form" for linear state eq'n

If state eq'n is linear, with invertible matrix E ,
then

$$\underline{\dot{Q}} = \underline{\hat{E}}(t) \underline{\dot{z}} + \underline{F}(t) \underline{z} + \underline{G}(t) \underline{u}$$

Can be re-written as

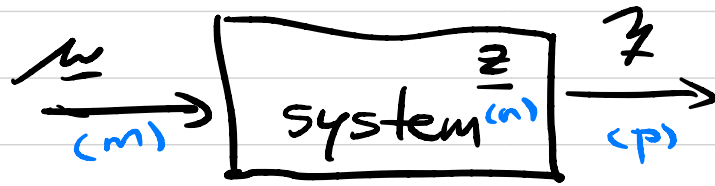
$$\underline{\dot{z}} = \underline{A}(t) \underline{z} + \underline{B}(t) \underline{u}$$

with

$$\underline{A}(t) = -\underline{E}(t)^{-1} \underline{F}(t)$$

$$\underline{B}(t) = -\underline{E}(t)^{-1} \underline{G}(t)$$

Note: $\underline{E}(t)$ will be invertible in all examples considered in this class.



$$\dot{\underline{z}} = \underline{f}(\underline{\dot{z}}, \underline{z}, \underline{u}, t) \quad \text{state eq'n}$$

$$\underline{y} = \underline{h}(\underline{z}, \underline{u}, t) \quad \text{output eq'n}$$

like state eq'n, output eq'n is linear, f.f

$$\underline{h}(\underline{z}, \underline{u}, t) = \underbrace{\underline{C}(t)}_{p \times n} \underline{z} + \underbrace{\underline{D}(t)}_{p \times m} \underline{u}$$

System is linear if both state and output eq'ns are

= Pendulum example (cart)

Suppose we want to control angle $\Theta = z$,
Then

$$\begin{aligned} \underline{y} = \underline{h}(\underline{z}, \underline{u}, t) &= \underline{z}, \\ &= \underbrace{[1 \quad 0]}_{\underline{C}} \underline{z} + \underbrace{0}_{\underline{D}} \underline{u} \end{aligned}$$

(linear and time-invariant here)

Why use $\sin z \approx z$ in pendulum example?

Recall $\sin z = z - \frac{z^3}{6} + \frac{z^5}{120} - \dots$ (H.O.T.)

and for $|z| \ll 1$, $z \gg z^3, z^5, \dots$ (H.O.T negligible)

Via Taylor, can do a similar expansion for most functions, around arbitrary reference points (other than 0)

$$f(z) = f(z^*) + f'(z^*)(z - z^*) + \frac{1}{2} f''(z^*)(z - z^*)^2 + \dots \text{ (HOT)}$$

so for small $|z - z^*|$, $f(z) - f(z^*) \approx f'(z^*)(z - z^*)$

similarly, if f depends on two (or more) variables

$$f(z_1, z_2) = f(z_1^*, z_2^*) + \left. \frac{\partial f}{\partial z_1} \right|_{z_1^*, z_2^*} (z_1 - z_1^*) + \left. \frac{\partial f}{\partial z_2} \right|_{z_1^*, z_2^*} (z_2 - z_2^*) + \dots \text{ (HOT)}$$

and we can do this with each function that appears in our problem ($f_1, f_2, \dots, f_n; h_1, h_2, \dots, h_p$)

Reference sol'n

Assume we know a sol'n $\underline{z}^*(t)$ for state eq'n given IC $\underline{z}^*(t_0)$ and input $\underline{u}^*(t)$

i.e.
$$\underline{0} = \underline{f}(\dot{\underline{z}}^*(t), \underline{z}^*(t), \underline{u}^*(t), t) \quad \forall t \geq t_0$$

any other sol'n $\underline{z}(t)$ must also satisfy

$$\underline{0} = \underline{f}(\dot{\underline{z}}(t), \underline{z}(t), \underline{u}(t), t) \quad \forall t \geq t_0$$

Differencing

$$\underline{0} = \underline{f}(\dot{\underline{z}}(t), \underline{z}(t), \underline{u}(t), t) - \underline{f}(\dot{\underline{z}}^*(t), \underline{z}^*(t), \underline{u}(t), t)$$

By Taylor

$$\underline{0} = E(t)[\dot{\underline{z}}(t) - \dot{\underline{z}}^*(t)] + F(t)[\underline{z}(t) - \underline{z}^*(t)] + G(t)[\underline{u}(t) - \underline{u}^*(t)]$$

+HOT

Where E, F, G are the Jacobian matrices

$$E_{ij} = \left\{ \frac{\partial f_i}{\partial \dot{z}_j} \right\}_* \quad i, j = 1, \dots, n \quad \left[E(t) = \frac{\partial f}{\partial \dot{z}} \Big|_{*}(t) \right]$$

$$F_{ij} = \left\{ \frac{\partial f_i}{\partial z_j} \right\}_* \quad i, j = 1, \dots, n \quad \left[F(t) = \frac{\partial f}{\partial z} \Big|_{*}(t) \right]$$

$$G_{ij} = \left\{ \frac{\partial f_i}{\partial w_j} \right\}_* \quad \begin{matrix} i = 1, \dots, n \\ j = 1, \dots, m \end{matrix} \quad \left[G(t) = \frac{\partial f}{\partial w} \Big|_{*}(t) \right]$$

where $\{ \cdot \}_*$ means the derivatives are evaluated
for the reference conditions

i.e. using $\dot{z} = \dot{z}^*(t)$, $z = z^*(t)$, $w = w^*(t)$

after taking the indicated derivatives.

Finally then we can write a model for deviations of the state vector from the assumed reference

$$\dot{\underline{x}}(t) = A(t)\underline{x}(t) + B(t)\underline{u}(t) + \underline{r}(t)$$

where

$$A(t) = -E(t)^{-1}F(t)$$

$$B(t) = -E(t)^{-1}G(t)$$

E, F, G jacobians of $\underline{f}$

and

$$\underline{x}(t) \triangleq \underline{z}(t) - \underline{z}^*(t)$$

Deviation from
ref. sol'n.

$$\underline{u}(t) \triangleq \underline{w}(t) - \underline{w}^*(t)$$

Deviation from
ref input

finally, $\underline{r}(t)$ has the H.O.T

so that $\|\underline{r}(t)\|$ small when $\|\underline{x}(t)\|$ and $\|\underline{u}(t)\|$ small

Important note:

$x(t) \rightarrow 0$ in linearized model
 $\Rightarrow z(t) \rightarrow z^*(t)$ in original (full) model
i.e. linearized stability $\Rightarrow$ full solution
converges back to reference sol'n
(Not to 0!!)

Also remember $u(t)$ is not the physical
input $w(t)$. It is the deviation of $w(t)$
from the reference input $w^*(t)$ that gives
rise to the ref sol'n $z^*(t)$ used in the
linearization.

Example - pendulum (cont)

$$f_1 = \dot{z}_1 - z_2$$

$$f_2 = ml^2 \dot{z}_2 + mgl \sin z_1 + b(z_2) - u$$

$$E(t) = \begin{bmatrix} \frac{\partial f_1}{\partial \dot{z}_1} & \frac{\partial f_1}{\partial \dot{z}_2} \\ \frac{\partial f_2}{\partial \dot{z}_1} & \frac{\partial f_2}{\partial \dot{z}_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & ml^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & ml^2 \end{bmatrix}$$

$$F'(t) = \begin{bmatrix} \frac{\partial f_1}{\partial z_1} & \frac{\partial f_1}{\partial z_2} \\ \frac{\partial f_2}{\partial z_1} & \frac{\partial f_2}{\partial z_2} \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ mgl \cos z_1 & b'(z_2) \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ mgl \cos z_1^*(t) & b'(z_2^*(t)) \end{bmatrix}$$

Depends on specific ref soln!

$$G(t) = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

If ref sol'n is $z_1^* = 0, z_2^* = 0, w^* = 0$

(pend is nominally motionless, straight down)

Then $F'(t) = \begin{bmatrix} 0 & -1 \\ mgl & b'(0) \end{bmatrix}$ (constant)

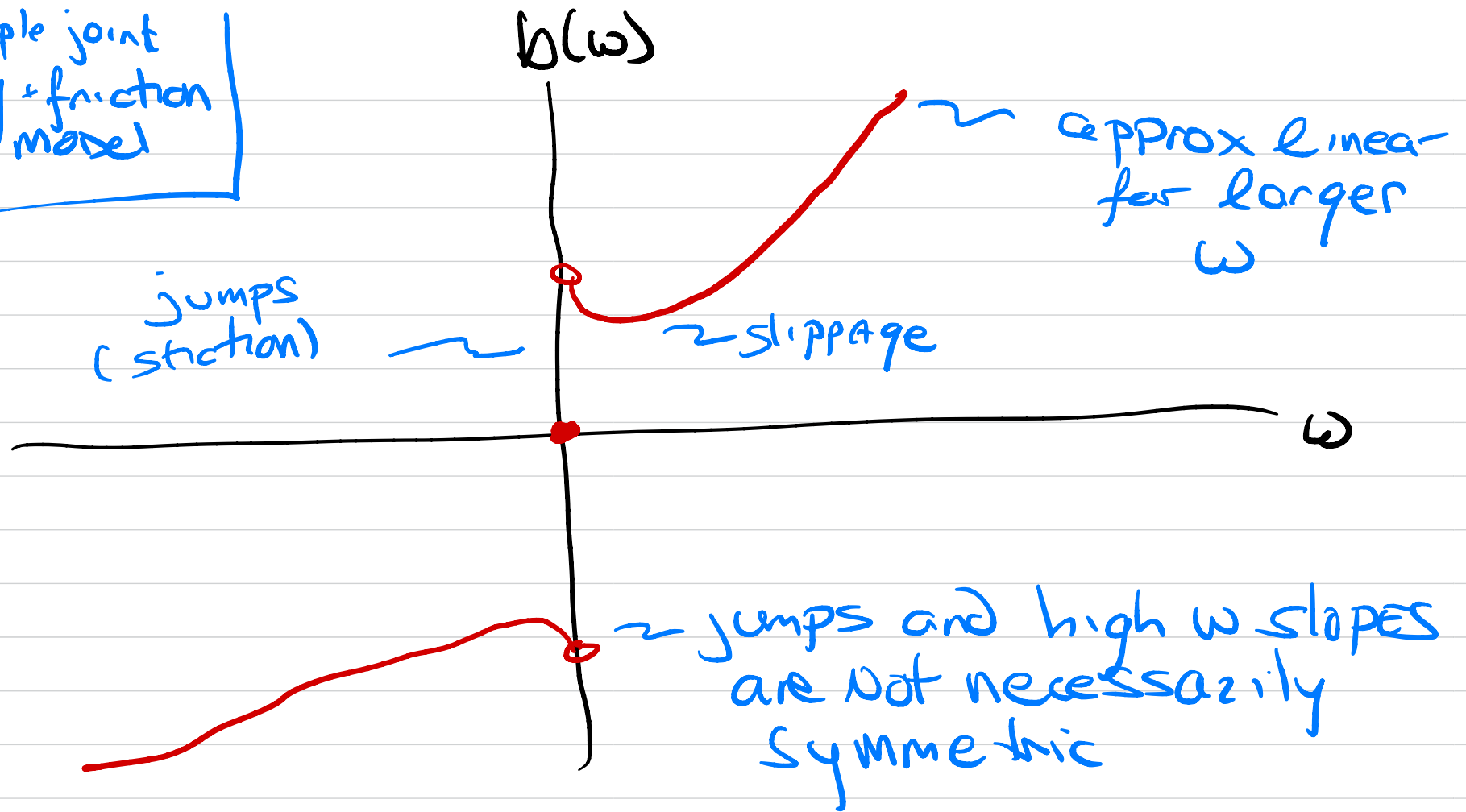
which agrees prev. hand-waving analysis.

Note $b'(0) = \langle \text{slope of friction curve at } 0 \rangle$

[Aside: friction is notoriously difficult to model accurately.

It is dissipative, so $b(\omega)\omega \geq 0$, but
can contain discontinuities:

Simple joint
drag + friction
model



Note: such a function cannot be completely treated within the formalism above since it fails to be differentiable at 0.

Finally note that we can similarly linearize the output eq'n for any reference sol'n

Let $y(t) = \underline{y}(t) - \underline{y}^*(t)$ where

$$\underline{y}^*(t) = \underline{h}(\underline{z}^*(t), \underline{u}^*(t)) \quad (\text{ref. output})$$

Then

$$\underline{y}(t) = C(t)\underline{x}(t) + D(t)\underline{u}(t) + \underline{\Gamma}_y(t)$$

where

$$C(t) = \left[\frac{\partial \underline{h}}{\partial \underline{z}} \right]_{\underline{z}^*(t)} \quad (\text{p} \times \text{n} \text{ jacobian of } \underline{h})$$

$$D(t) = \left[\frac{\partial \underline{h}}{\partial \underline{u}} \right]_{\underline{u}^*(t)} \quad (\text{p} \times \text{m} \text{ jacobian of } \underline{h})$$

and $\underline{\Gamma}_y(t)$ has H.O.T which are negligible if $\|\underline{z}(t)\|$ and $\|\underline{u}(t)\|$ are small.

Note Again: $y(t)$ is deviation of output from reference, not physical output.

Above $\dot{\underline{z}}^* = \underline{0}$ case

is representative of a special category of reference sol'n's known as "equilibria". Such sol'n's are defined by $\dot{\underline{z}}^* = \underline{0} \Rightarrow \underline{z}^* = \underline{\text{constant}}$

$$\Rightarrow \underline{0} = \underline{f}(\underline{0}, \underline{z}^*, \underline{w}^*) \quad \left[\begin{array}{l} \text{state eq'n, w/ above} \\ \text{restrictions} \end{array} \right]$$

This is no longer a DE, but rather a system of nonlinear equations which can be solved (analytically or numerically) for $\underline{z}^*$ given $\underline{w}^*$, (or alternately for $\underline{w}^*$ given $\underline{z}^*$.)

In a/c applications these equations are often referred to as the "Trim conditions"

For pendulum, equilibria satisfy

$$0 = \cancel{\dot{z}_1^*} - z_2^* \Rightarrow z_2^* = 0$$

$$0 = ml^2 \cancel{\dot{z}_2^*} + mgl \sin z_1^* + \underbrace{b(\cancel{z_2^*})}_{b(0)=0} - \tau^*$$

$$\Rightarrow \boxed{mgl \sin z_1^* = \tau^*}$$

$$\text{or } z_1^* = \sin^{-1}\left(\frac{\tau^*}{mgl}\right)$$

where τ^* is the "trim" torque applied at pivot

$$\text{i.e. } \tau^* = 0 \Rightarrow z_1^* = 0$$

$$\tau^* = \frac{1}{2}mgl \Rightarrow z_1^* = \pi/6 \quad (30^\circ)$$

$$\tau^* = \frac{\sqrt{2}}{2}mgl \Rightarrow z_1^* = \pi/4 \quad (45^\circ)$$

$$\tau^* = mgl \Rightarrow z_1^* = \pi/2 \quad (90^\circ) \text{ etc.}$$

For these equilibria, E and G in linearization are unchanged, but

$$F = \begin{bmatrix} 0 & -1 \\ mgl \cos z_i^* & b'(0) \end{bmatrix}$$

↑ changes depending on specific equilibrium used in linearization!

How do these changes affect properties of corresponding linearized model?